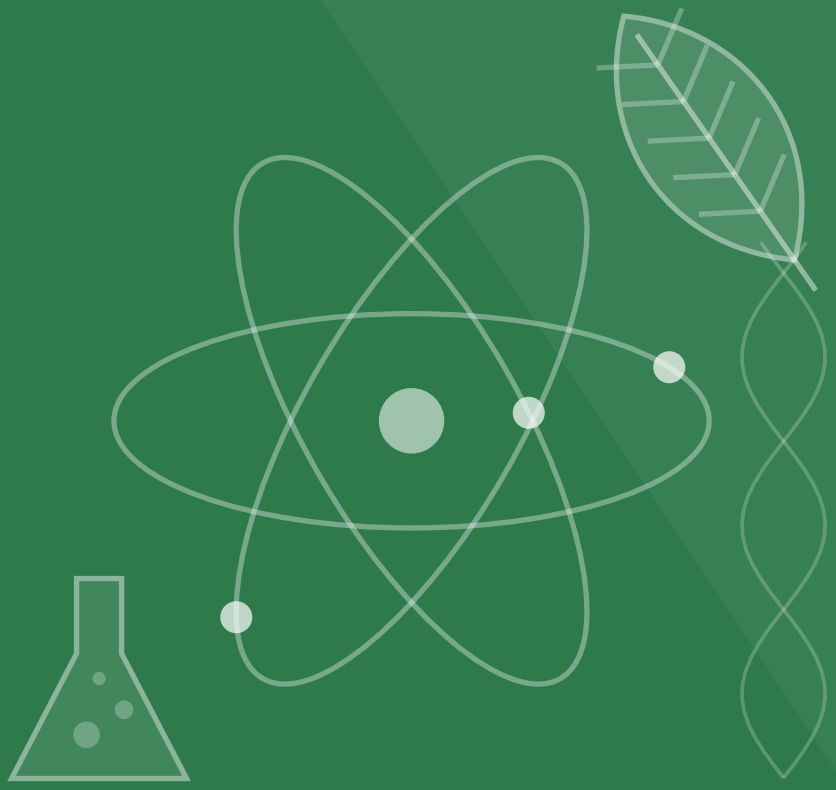
VENVLEARNING

STUDENT TEXTBOOK



Science

GRADE 4

Lessons, examples, and practice for the whole year

2026–2027 School Year · Unit 1: The Human Body

VENVLEARNING

Grade 4
Science

STUDENT TEXTBOOK

2026–2027 School Year

Unit 1 The Human Body

This book belongs to _____

A VenvLearning family homeschool curriculum

How to use this book

Each lesson starts with an I can goal and a I Wonder question. Read the lesson with your teacher. Read the Explain It boxes to see how a scientist reasons from evidence, and answer the Check Your Understanding questions. Your practice pages are in your Student Workbook, labeled with the same lesson number.

- Key Idea boxes hold the most important rule or fact. Learn these.
- Watch Out boxes warn you about common mistakes.
- Investigate boxes are hands-on experiments. Do them with your teacher and record what you observe.
- Every Friday there is a short Weekly Check in your workbook. The last Friday of a unit is the Unit Test.

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UNIT 1

The Human Body

Thinking like a scientist; cells, tissues, and organs; and the body systems that keep you alive.

Your body is an amazing machine made of trillions of cells. In this unit you'll learn to think like a scientist: observing, asking testable questions, running fair tests, and using evidence. Then you'll use those skills to explore how your body is organized and how its systems work together.

In this unit you will…

- Tell observations from inferences, and plan fair tests.
- Support claims with evidence and reasoning (CER).
- Describe levels of organization from cells to systems.
- Explain the jobs of the digestive, respiratory, circulatory, skeletal, muscular, and nervous systems and how they work together.

SCIENCE IN YOUR WORLD

Doctors, nurses, athletes, and coaches use what they know about body systems every day to keep people healthy.

UNIT VOCABULARY

observation	Information gathered with senses or tools.
inference	A conclusion based on observations.
fair test	Only one variable is changed.
cell	The smallest living unit.
organ system	Organs that work together.
nutrients	Substances in food the body uses.

Lessons in this unit

Lesson	Title
1.1	Thinking Like a Scientist

Lesson	Title
1.2	Fair Tests and Evidence
1.3	Cells, Tissues, Organs, and Systems
1.4	Body Systems Work Together
1.5	The Digestive System
1.6	The Respiratory System
1.7	The Heart and Blood Vessels
1.8	Bones, Muscles, and Healthy Choices

UNIT 1 · LESSON 1.1 · Science and Engineering Practices 1, 3

Lesson 1.1: Thinking Like a Scientist

I can tell the difference between an observation and an inference, and ask a question that can be investigated.

WORDS TO KNOW

observation	Information you gather with your senses or with tools.
inference	A conclusion you reach from observations and what you already know.
investigate	To find out about something by observing, testing, and measuring.

I WONDER

You walk into the kitchen and see crumbs on the counter and an empty cookie jar. What do you SEE? What do you THINK happened?

Scientists are curious people who ask questions about the world and then look for evidence to answer them. Two of their most important tools are observations and inferences.

Observation	Inference
what you notice with your senses or tools	what you figure out from what you noticed
“The cookie jar is empty.”	“Someone ate the cookies.”
“The ground is wet.”	“It rained last night.”
“The thermometer reads 38°F.”	“It is too cold to swim outside.”

An inference can be wrong. Maybe the cookies were moved to another jar! Good scientists gather more observations to test their inferences.

EXPLAIN IT · ASK A TESTABLE QUESTION

You notice that a puddle in the sun dries faster than a puddle in the shade.

1. Observation: the sunny puddle disappeared first.
2. Question that can't be tested easily: “Why does water exist?”
3. Testable question: “Does water evaporate faster in sunlight than in shade?” You can set up two cups of water and measure.

Explanation: A testable question can be answered by observing, measuring, or experimenting.

INVESTIGATE · MYSTERY BAG

What can you figure out about an object you can't see?

You need: a paper bag with a hidden object (chosen by your teacher); notebook.

1. Without looking, shake the bag and listen. Feel the object through the paper.
2. List at least four observations (sound, weight, shape, texture).
3. Write one inference about what the object is, and the observations that support it.
4. Open the bag. Was your inference correct?

Record: a two-column chart: Observations | Inferences.

KEY IDEA

Observations come from your senses and tools. Inferences are explanations based on observations. Testable questions can be answered with evidence.

CHECK YOUR UNDERSTANDING

Answer in your science notebook in complete sentences. Then check with your teacher.

1. Observation or inference: “The leaf is green and has jagged edges.”
2. Observation or inference: “The dog is hungry because it is barking at its bowl.”
3. Make this question testable: “Why do plants like light?”

More practice: Workbook, Lesson 1.1.

LESSON SUMMARY

Observations come from senses and tools; inferences explain them; testable questions can be answered with evidence.

UNIT 1 · LESSON 1.2 · Science and Engineering Practices 3, 4, 7

Lesson 1.2: Fair Tests and Evidence

I can plan a fair test and use evidence to support a claim.

WORDS TO KNOW

variable	Something that can change in an investigation.
fair test	An investigation where only one variable is changed on purpose.
evidence	Observations and data that support a claim.
claim	A statement that answers a question.

I WONDER

Two friends race paper airplanes. One throws hard and one throws gently. Is it a fair race? Why or why not?

In a fair test, you change only one thing, the variable you test, and keep everything else the same. Then you measure what happens. If you change two things at once, you can't tell which one caused the result.

Question	Change (test)	Keep the same	Measure
Which paper towel brand absorbs the most water?	brand of towel	towel size, amount of water, time	water left in the cup (mL)

After a test, scientists write a claim, support it with evidence (data), and give reasoning that explains why the evidence supports the claim. This is called CER.

Claim: answers the question → Evidence: data that supports it → Reasoning: explains why the evidence supports the claim

INVESTIGATE · THE PAPER TOWEL TEST

Which paper towel absorbs the most water?

You need: 2 or 3 kinds of paper towel cut to the same size; measuring cup; water; 3 cups.

1. Pour 100 mL of water into each cup.
2. Dip one towel square in each cup for 10 seconds.
3. Lift the towel and let it drip back into the cup for 10 seconds.
4. Measure the water left in each cup.

Record: a data table, then a CER: Which towel absorbed the most? What is your evidence?

KEY IDEA

Change one variable, keep the rest the same, measure the result, and support your claim with evidence and reasoning.

CHECK YOUR UNDERSTANDING

Answer in your science notebook in complete sentences. Then check with your teacher.

1. Why must the towel squares be the same size?
2. What is the claim in CER?
3. A student tests plant growth but gives one plant more water AND more light. What is wrong?

More practice: Workbook, Lesson 1.2.

LESSON SUMMARY

A fair test changes one variable; CER = claim, evidence, reasoning.

UNIT 1 · LESSON 1.3 · 4-LS1-1

Lesson 1.3: Cells, Tissues, Organs, and Systems

I can describe how the body is organized from cells to organ systems.

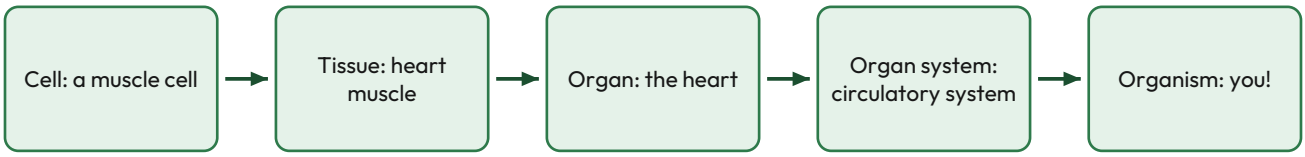
WORDS TO KNOW

cell	The smallest living unit of a living thing.
tissue	A group of similar cells that work together.
organ	A body part made of tissues that does a special job.
organ system	A group of organs that work together.

I WONDER

A brick is part of a wall, a wall is part of a room, and a room is part of a house. How might your body be built the same way?

Your body is made of trillions of tiny cells, so small you need a microscope to see them. Similar cells work together as tissues, such as muscle tissue. Different tissues form organs, like the heart or stomach. Organs work together in organ systems.



There are many kinds of cells. Nerve cells are long and thin to carry messages. Red blood cells are round and flat to carry oxygen. A cell's shape fits its job.

INVESTIGATE · BUILD THE LEVELS

How can a model show levels of organization?

You need: building blocks or LEGO bricks; sticky notes.

1. Let one brick stand for one cell.
2. Join several matching bricks into a "tissue."
3. Combine different "tissues" into an "organ."
4. Put several organs together into a "system." Label each level with a sticky note.

Record: a sketch of your model with each level labeled.

KEY IDEA

Cells → tissues → organs → organ systems → organism.

CHECK YOUR UNDERSTANDING

Answer in your science notebook in complete sentences. Then check with your teacher.

1. What is the smallest living unit of the body?
2. Put in order: organ, cell, organ system, tissue
3. Name one organ.

More practice: Workbook, Lesson 1.3.

LESSON SUMMARY

Cells form tissues, tissues form organs, organs form systems.

UNIT 1 · LESSON 1.4 · 4-LS1-1

Lesson 1.4: Body Systems Work Together

I can name major body systems, describe each one's main job, and explain how they work together.

WORDS TO KNOW

skeletal system	Bones that support and protect the body.
muscular system	Muscles that move the body.
nervous system	The brain, spinal cord, and nerves that control the body.
digestive system	Organs that break down food.
respiratory system	Organs that bring in oxygen and remove carbon dioxide.
circulatory system	The heart and blood vessels that move blood.

I WONDER

Think about kicking a soccer ball. Which parts of your body are working?

System	Main organs	Main job
skeletal	bones	supports, protects, works with muscles to move
muscular	muscles	moves bones and organs
nervous	brain, spinal cord, nerves	senses, thinks, sends messages
digestive	mouth, stomach, intestines	breaks food into nutrients
respiratory	nose, windpipe, lungs	takes in oxygen, gets rid of carbon dioxide
circulatory	heart, blood vessels, blood	carries oxygen and nutrients to cells

EXPLAIN IT · SYSTEMS WORKING TOGETHER

Kicking a soccer ball

1. Nervous: your eyes see the ball and your brain sends a message to your leg.
2. Muscular and skeletal: leg muscles pull on bones to swing your leg.
3. Respiratory and circulatory: you breathe faster and your heart pumps faster to deliver oxygen to your muscles.
4. Digestive: nutrients from food give your cells energy.

Explanation: No system works alone; all of them work together to keep you alive and moving.

INVESTIGATE · SYSTEMS RELAY

Which systems work when you do an everyday task?

You need: index cards labeled with the six systems; a timer.

1 Choose an action: eating an apple, running up stairs, or reading a book aloud.

2 Do the action (safely!).

3 Hold up every system card you think was working, and explain why.

Record: the action and the systems used, with one reason for each.

KEY IDEA

Each body system has a main job, and the systems depend on one another.

CHECK YOUR UNDERSTANDING

Answer in your science notebook in complete sentences. Then check with your teacher.

1. Which system carries oxygen to cells?
2. Which system breaks down food?
3. Which systems help you jump?

More practice: Workbook, Lesson 1.4.

LESSON SUMMARY

Six major systems each have a job, and they work together.

UNIT 1 · LESSON 1.5 · 4-LS1-1

Lesson 1.5: The Digestive System

I can describe the path of food through the digestive system and explain what each organ does.

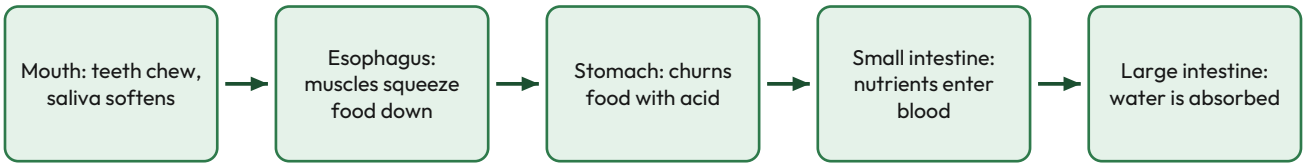
WORDS TO KNOW

nutrients	Substances in food that the body uses for energy, growth, and repair.
esophagus	The tube that carries food from the mouth to the stomach.
small intestine	The long, coiled tube where most nutrients pass into the blood.
large intestine	The wider tube where water is absorbed from leftover food.

I WONDER

Where does your breakfast go after you swallow it?

Your body can't use a sandwich as a sandwich. The digestive system breaks food into tiny nutrients that can pass into your blood and travel to your cells.



Digestion is both mechanical (chewing and churning break food into smaller pieces) and chemical (saliva, stomach acid, and other juices break food down into nutrients). The liver and pancreas make juices that help in the small intestine. In an adult, the small intestine is about 20 feet long, coiled up to fit inside the body!

INVESTIGATE · STOMACH IN A BAG

How does the stomach help break down food?

You need: zip-top plastic bag; 4 crackers; 1/2 cup water; a few drops of vinegar (to stand for acid).

1. Put the crackers in the bag. Observe them.
2. Add the water and vinegar and seal the bag tightly.
3. Gently squeeze and mash the bag for 2 minutes, the way stomach muscles churn.
4. Observe the crackers again.

Record: before-and-after observations, and which part of digestion the squeezing models.

Safety: Do not eat or drink anything used in an investigation.

KEY IDEA

Food travels mouth → esophagus → stomach → small intestine → large intestine. Nutrients enter the blood in the small intestine.

CHECK YOUR UNDERSTANDING

Answer in your science notebook in complete sentences. Then check with your teacher.

1. Where do most nutrients enter the blood?
2. What does the large intestine absorb?
3. What did squeezing the bag model?

More practice: Workbook, Lesson 1.5.

LESSON SUMMARY

Digestion breaks food into nutrients that enter the blood in the small intestine.

UNIT 1 · LESSON 1.6 · 4-LS1-1

Lesson 1.6: The Respiratory System

I can explain how the respiratory system brings oxygen into the body and removes carbon dioxide.

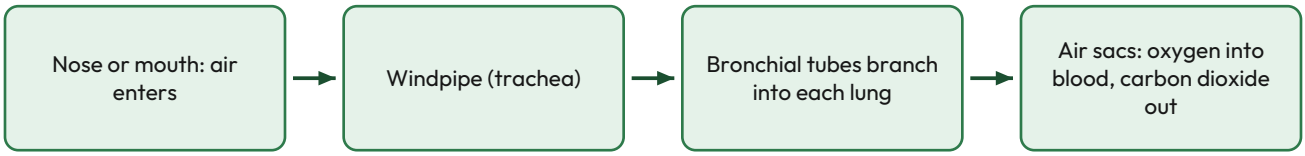
WORDS TO KNOW

oxygen	A gas in air that cells need to get energy from food.
carbon dioxide	A waste gas your cells make and you breathe out.
diaphragm	A dome-shaped muscle below the lungs that helps you breathe.
air sacs	Tiny balloon-like sacs in the lungs where gases pass into and out of the blood.

I WONDER

Hold your breath for a few seconds. Why can't you hold it forever?

Every cell in your body needs oxygen to release energy from nutrients. Cells also make carbon dioxide, a waste gas. The respiratory system brings oxygen in and sends carbon dioxide out.



When the diaphragm tightens and moves down, the chest gets bigger and air rushes into the lungs. When it relaxes and moves up, air is pushed out. Inside the lungs are millions of tiny air

sacs wrapped in blood vessels, where oxygen moves into the blood and carbon dioxide moves out.

INVESTIGATE · BREATHING RATE

How does exercise change how fast you breathe?

You need: a timer; a partner.

1. Sit quietly for 1 minute. Count your breaths for 30 seconds and multiply by 2.
2. Do jumping jacks for 1 minute.
3. Right away, count your breaths for 30 seconds and multiply by 2.
4. Rest 3 minutes and count again.

Record: breaths per minute at rest, after exercise, and after resting, in a table and a bar graph.

KEY IDEA

You breathe in oxygen that cells need and breathe out carbon dioxide that cells make. Gases are exchanged in the air sacs.

CHECK YOUR UNDERSTANDING

Answer in your science notebook in complete sentences. Then check with your teacher.

1. What gas do your cells need?
2. What muscle helps you breathe?
3. Why does breathing speed up during exercise?

More practice: Workbook, Lesson 1.6.

LESSON SUMMARY

The respiratory system takes in oxygen and removes carbon dioxide.

UNIT 1 · LESSON 1.7 · 4-LS1-1

Lesson 1.7: The Heart and Blood Vessels

I can describe how the heart and blood vessels move blood through the body, and measure my pulse.

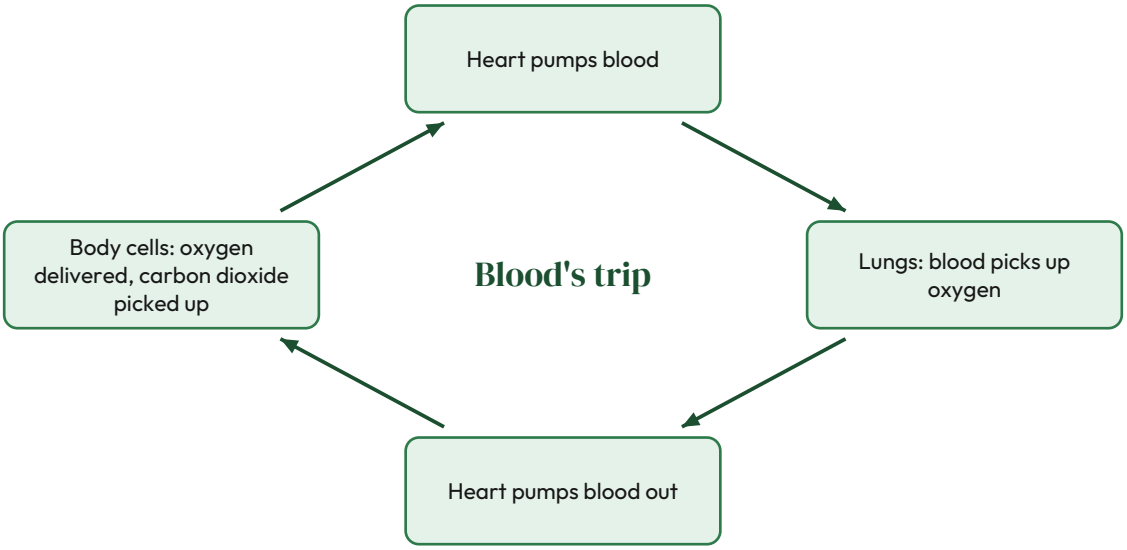
WORDS TO KNOW

heart	A muscular organ that pumps blood.
artery	A blood vessel that carries blood away from the heart.
vein	A blood vessel that carries blood back to the heart.
capillary	A tiny blood vessel where materials pass between blood and cells.
pulse	The beat you can feel as the heart pushes blood through an artery.

I WONDER

Make a fist. Your heart is about that size. How many times do you think it beats in one minute?

The heart is a muscle about the size of your fist. It has four chambers and pumps blood through a huge network of blood vessels. Arteries carry blood away from the heart. Capillaries are tiny vessels where oxygen and nutrients move into cells and wastes move out. Veins carry blood back to the heart.



INVESTIGATE · TAKE YOUR PULSE

How does activity change heart rate?

You need: a timer.

1 Place two fingers (not your thumb) on the side of your neck or the inside of your wrist.

2 Count the beats for 15 seconds. Multiply by 4 to find beats per minute.

3 Run in place for 1 minute, then measure again.

4 Rest 3 minutes and measure again.

Record: beats per minute in a table; compare with your breathing data from Lesson 1.6.

WATCH OUT

Blood in veins is not blue. Blood is always red; veins can look bluish through skin because of how light passes through it.

KEY IDEA

The heart pumps blood through arteries to capillaries and back through veins, delivering oxygen and nutrients and carrying away wastes.

CHECK YOUR UNDERSTANDING

Answer in your science notebook in complete sentences. Then check with your teacher.

1. Which vessels carry blood away from the heart?
2. Where do oxygen and nutrients move into cells?
3. 15-second count is 22 beats. Beats per minute?

More practice: Workbook, Lesson 1.7.

LESSON SUMMARY

The heart pumps blood through arteries, capillaries, and veins.

UNIT 1 · LESSON 1.8 · 4-LS1-1

Lesson 1.8: Bones, Muscles, and Healthy Choices

I can explain how bones and muscles work together and describe choices that keep body systems healthy.

WORDS TO KNOW

joint	A place where two bones meet.
tendon	A tough cord that connects a muscle to a bone.
ligament	A tough band that connects bone to bone.

I WONDER

Bend and straighten your arm. What do you feel happening in your upper arm?

An adult skeleton has 206 bones. Bones support the body, protect organs (the skull protects the brain; the ribs protect the heart and lungs), and make blood cells inside. Muscles are attached to bones by tendons. Muscles can only pull, not push, so they work in pairs.

EXPLAIN IT · MUSCLE PAIRS

Bending and straightening your arm

1. To bend: the biceps (front of upper arm) tightens and pulls the forearm up; the triceps relaxes.
2. To straighten: the triceps (back of upper arm) tightens and pulls; the biceps relaxes.

Explanation: Muscles work in pairs because each muscle can only pull.

Healthy choice	Systems it helps
active play about 60 minutes a day	muscular, skeletal, circulatory, respiratory
foods like fruits, vegetables, whole grains, and protein	digestive (and every system that needs nutrients)
9–12 hours of sleep a night for children your age	nervous (and all systems repair during sleep)
drinking water	digestive, circulatory
wearing a helmet and seat belt	skeletal, nervous

INVESTIGATE · MUSCLE PARTNERS

How do muscles work in pairs?

You need: your own arm.

1 Place your hand on your biceps. Slowly bend your arm. What does the muscle do?

2 Move your hand to your triceps. Straighten your arm. What does that muscle do?

Record: a labeled sketch of the upper arm showing which muscle pulls to bend and which pulls to straighten.

KEY IDEA

Bones support and protect; muscles pull in pairs to move bones at joints. Healthy habits keep all systems working.

CHECK YOUR UNDERSTANDING

Answer in your science notebook in complete sentences. Then check with your teacher.

1. How many bones in an adult skeleton?
2. What connects muscle to bone?
3. Why do muscles work in pairs?

More practice: Workbook, Lesson 1.8.

LESSON SUMMARY

Bones support and protect; muscles pull in pairs; healthy habits keep systems strong.

UNIT 1 REVIEW

Unit 1 Study Guide

Use this page to get ready for the unit test. Read each big idea, then try the review pages in your workbook.

Lesson	Remember…
1.1	Observations come from senses and tools; inferences explain them; testable questions can be answered with evidence.
1.2	A fair test changes one variable; CER = claim, evidence, reasoning.
1.3	Cells form tissues, tissues form organs, organs form systems.
1.4	Six major systems each have a job, and they work together.
1.5	Digestion breaks food into nutrients that enter the blood in the small intestine.
1.6	The respiratory system takes in oxygen and removes carbon dioxide.
1.7	The heart pumps blood through arteries, capillaries, and veins.
1.8	Bones support and protect; muscles pull in pairs; healthy habits keep systems strong.

Glossary

air sacs	Tiny balloon-like sacs in the lungs where gases pass into and out of the blood.
artery	A blood vessel that carries blood away from the heart.
capillary	A tiny blood vessel where materials pass between blood and cells.
carbon dioxide	A waste gas your cells make and you breathe out.
cell	The smallest living unit.
circulatory system	The heart and blood vessels that move blood.
claim	A statement that answers a question.
diaphragm	A dome-shaped muscle below the lungs that helps you breathe.
digestive system	Organs that break down food.
esophagus	The tube that carries food from the mouth to the stomach.
evidence	Observations and data that support a claim.
fair test	Only one variable is changed.
heart	A muscular organ that pumps blood.
inference	A conclusion based on observations.
investigate	To find out about something by observing, testing, and measuring.
joint	A place where two bones meet.
large intestine	The wider tube where water is absorbed from leftover food.
ligament	A tough band that connects bone to bone.
muscular system	Muscles that move the body.
nervous system	The brain, spinal cord, and nerves that control the body.
nutrients	Substances in food the body uses.
observation	Information gathered with senses or tools.
organ	A body part made of tissues that does a special job.
organ system	Organs that work together.
oxygen	A gas in air that cells need to get energy from food.
pulse	The beat you can feel as the heart pushes blood through an artery.
respiratory system	Organs that bring in oxygen and remove carbon dioxide.
skeletal system	Bones that support and protect the body.

small intestine	The long, coiled tube where most nutrients pass into the blood.
tendon	A tough cord that connects a muscle to a bone.
tissue	A group of similar cells that work together.
variable	Something that can change in an investigation.
vein	A blood vessel that carries blood back to the heart.

Grade 4 Science

Student Textbook · 2026–2027

A full year of Grade 4 science built around questions, investigations, and evidence. Students observe, measure, model, and explain the natural world, then use what they learn to design solutions to real problems.

This Student Textbook teaches every lesson: the big idea, key vocabulary, models and examples, and a short set of practice problems to check understanding. It pairs with the Student Workbook and the Teacher Edition.

INSIDE THIS YEAR

01 The Human Body

IN THIS BOOK

- Clear lesson goals: “I can…” statements
- Key vocabulary with student-friendly definitions
- Models, examples, and Key Idea summaries
- A study guide and glossary for every unit

THE COMPLETE SET



Textbook
You are holding this book

Workbook

Teacher Edition

Each lesson has the same number in all three books, and every day of the year is mapped to its lesson in the VenvLearning planner.